

Reward Function Ablation Study: Hierarchical RL for Crane Log Grasping

Crane Testbed Project

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1 Overview

We conducted a 2×2 factorial ablation study comparing reward function designs for hierarchical reinforcement learning in crane log grasping. The two factors are:

1. **Formula Type:** Multiplicative vs. Additive combination of efficiency and alignment
2. **Normalization:** Raw log count vs. Normalized efficiency (logs grasped / logs available)

Additionally, each variant was tested with and without **Domain Randomization (DR)** (variable pile sizes 50–200 logs).

2 Grasp Evaluation Metrics

2.1 Logs Grasped (n)

After the grapple lifts (LIFT_HIGH phase), we count logs within a proximity radius $r_{\text{prox}} = 1.5\text{m}$ of the basegrapple center:

$$n = \left| \left\{ \log_i : \|\mathbf{p}_{\log_i} - \mathbf{p}_{\text{bg}}\| < r_{\text{prox}} \right\} \right| \quad (1)$$

where $\mathbf{p}_{\text{bg}}$ is the basegrapple position and $\mathbf{p}_{\log_i}$ is each log’s position in world frame.

2.2 Orientation Alignment (a)

Alignment measures how well the grasped logs are oriented relative to the grapple. For each grasped log, we compute the dot product between forward directions:

$$a_i = \left| \hat{\mathbf{f}}_{\text{bg}} \cdot \hat{\mathbf{f}}_{\log_i} \right| \quad (2)$$

where:

- $\hat{\mathbf{f}}_{\text{bg}} = R(\mathbf{q}_{\text{bg}}) \cdot [1, 0, 0]^T$ is the basegrapple’s forward direction (rotated from body frame)
- $\hat{\mathbf{f}}_{\log_i} = R(\mathbf{q}_{\log_i}) \cdot [1, 0, 0]^T$ is the log’s forward direction (along its length axis)
- $R(\mathbf{q})$ converts quaternion to rotation matrix
- Absolute value handles logs pointing in either direction along their axis

The average alignment over all grasped logs:

$$a = \frac{1}{n} \sum_{i=1}^n a_i \in [0, 1] \quad (3)$$

Interpretation:

- $a = 1.0$: All logs perfectly parallel to grapple (ideal for clean lift)
- $a = 0.0$: All logs perpendicular to grapple (will slip or disturb pile)
- $a > 0.7$: Good alignment threshold—logs lift cleanly without disrupting pile

2.3 Normalized Efficiency (η)

For pile-size agnostic rewards, we normalize by logs available at the target position:

$$\eta = \frac{n}{\min(\max(n_{\text{avail}}, 1), n_{\text{max}})} \in [0, 1] \quad (4)$$

where:

- n_{avail} = logs within a vertical slice of the rack at the grapple’s Y position (action bounds prism)
- $n_{\text{max}} = 20$ = physical grapple capacity (prevents $\eta > 1$ for sparse areas)
- The $\max(\cdot, 1)$ prevents division by zero

3 Reward Function Variants

3.1 Alignment Bonus (Multiplicative Variants Only)

The alignment bonus rewards grasps with $a > 0.7$ (good grapple-log alignment):

$$b_{\text{align}} = \begin{cases} \text{efficiency} \times (a - 0.7) \times 5 & \text{if } a > 0.7 \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

This creates a $6\times$ steeper gradient above $a = 0.7$, strongly incentivizing good alignment.

3.2 Variant MR: Multiplicative Raw (Original Baseline)

$$r_{\text{MR}} = \begin{cases} n \cdot a + n \cdot (a - 0.7) \cdot 5 & \text{if } n > 0 \text{ and } a > 0.7 \\ n \cdot a & \text{if } n > 0 \text{ and } a \leq 0.7 \\ -3.0 & \text{if } n = 0 \text{ (failed grasp)} \end{cases} \quad (6)$$

Simplified form (for $a > 0.7$):

$$r_{\text{MR}} = n \cdot (6a - 3.5) \quad (7)$$

Example calculations:

- 10 logs, $a = 0.5$: $r = 10 \times 0.5 = 5.0$

- 10 logs, $a = 0.8$: $r = 10 \times 0.8 + 10 \times 0.1 \times 5 = 8 + 5 = 13.0$
- 20 logs, $a = 0.9$: $r = 20 \times 0.9 + 20 \times 0.2 \times 5 = 18 + 20 = 38.0$

Properties:

- Reward scales with absolute log count (favors larger piles)
- Strong coupling: low alignment zeros out efficiency gains
- Range: $[-3, \sim 150]$ (depends on pile size)

3.3 Variant MN: Multiplicative Normalized

$$r_{\text{MN}} = \begin{cases} \eta \cdot 10 \cdot a + \eta \cdot 10 \cdot (a - 0.7) \cdot 5 & \text{if } n > 0 \text{ and } a > 0.7 \\ \eta \cdot 10 \cdot a & \text{if } n > 0 \text{ and } a \leq 0.7 \\ -3.0 & \text{if } n = 0 \text{ (failed grasp)} \end{cases} \quad (8)$$

Example calculations (assuming $n_{\text{avail}} = 20$):

- 10 logs, $a = 0.5$: $\eta = 0.5$, $r = 0.5 \times 10 \times 0.5 = 2.5$
- 10 logs, $a = 0.8$: $\eta = 0.5$, $r = 4.0 + 0.5 \times 10 \times 0.1 \times 5 = 4 + 2.5 = 6.5$
- 20 logs, $a = 0.9$: $\eta = 1.0$, $r = 9.0 + 1.0 \times 10 \times 0.2 \times 5 = 9 + 10 = 19.0$

Properties:

- Scale factor of 10 maintains similar magnitude to MR for typical grasps
- Pile-size agnostic: 50% efficiency on sparse pile = 50% on dense pile
- Range: $[-3, 25]$ (bounded, consistent across pile sizes)

3.4 Variant AR: Additive Raw

$$r_{\text{AR}} = \begin{cases} \frac{n}{20} + a & \text{if } n > 0 \\ -0.5 & \text{if } n = 0 \text{ (failed grasp)} \end{cases} \quad (9)$$

Example calculations:

- 10 logs, $a = 0.5$: $r = 0.5 + 0.5 = 1.0$
- 10 logs, $a = 0.8$: $r = 0.5 + 0.8 = 1.3$
- 20 logs, $a = 0.9$: $r = 1.0 + 0.9 = 1.9$

Properties:

- Decoupled: efficiency and alignment contribute independently
- No alignment bonus discontinuity (smooth reward landscape)
- Smaller failure penalty (-0.5 vs -3.0)—less punishing early exploration
- Range: $[-0.5, \sim 11]$

3.5 Variant AN: Additive Normalized

$$r_{\text{AN}} = \begin{cases} \eta + a & \text{if } n > 0 \\ -0.5 & \text{if } n = 0 \text{ (failed grasp)} \end{cases} \quad (10)$$

Example calculations:

- 10/20 logs, $a = 0.5$: $\eta = 0.5$, $r = 0.5 + 0.5 = 1.0$
- 10/20 logs, $a = 0.8$: $\eta = 0.5$, $r = 0.5 + 0.8 = 1.3$
- 20/20 logs, $a = 0.9$: $\eta = 1.0$, $r = 1.0 + 0.9 = 1.9$

Properties:

- Both terms in $[0, 1]$, sum in $[0, 2]$
- Clean interpretation: “grasp quality = efficiency + alignment”
- Fully pile-size agnostic
- Tightest bounded range: $[-0.5, 2.0]$

4 Summary Comparison

Table 1: Reward Function Variant Summary

Variant	Formula	Norm	Align Bonus	Fail Penalty	Range
MR	$n \cdot a$	Raw	$n(a-0.7) \cdot 5$	-3.0	$[-3, \sim 150]$
MN	$\eta \cdot 10 \cdot a$	Norm	$\eta \cdot 10(a-0.7) \cdot 5$	-3.0	$[-3, 25]$
AR	$\frac{n}{20} + a$	Raw	None	-0.5	$[-0.5, \sim 11]$
AN	$\eta + a$	Norm	None	-0.5	$[-0.5, 2]$

5 Training Results

5.1 Without Domain Randomization

Table 2: Training Performance (Fixed 200-log piles)

Variant	Iters	Final	Max	@25%	@50%	Converged?
MR (baseline)	505	99.51	99.68	90.41	88.21	Yes
MN	304	70.21	71.36	47.94	52.87	Improving
AR	196	17.18	17.18	15.20	15.99	Improving
AN	284	22.33	22.59	18.61	20.23	Improving

Note: Absolute reward values are **not directly comparable** across formula types due to different scales. MR achieving 99.51 and AN achieving 22.33 does not mean MR is $4\times$ better—they use different reward functions.

5.2 With Domain Randomization

Table 3: Training Performance (Variable 50–200 log piles)

Variant	Iters	Final	Max	@25%	@50%	Trend
DR-MN	197	44.90	48.84	16.59	39.82	$\uparrow\uparrow$ (+171%)
DR-AR	244	7.25	8.76	5.32	6.97	$\uparrow$ (+36%)
DR-AN	345	13.82	14.44	12.27	13.11	$\rightarrow$ (+13%)

6 Key Observations

6.1 Multiplicative vs. Additive

- **Multiplicative** (MR, MN) shows faster convergence and stronger learning signal
- The alignment bonus at $a > 0.7$ creates steep gradient toward good alignment
- **Additive** (AR, AN) provides smoother reward landscape without discontinuity
- Additive decouples efficiency/alignment—each component provides independent gradient

6.2 Normalized vs. Raw

- **Normalization** essential for domain randomization (consistent scale across pile sizes)
- Raw rewards bias toward larger piles (more logs $\Rightarrow$ higher reward regardless of efficiency)
- MR baseline achieves high rewards partly because 200-log piles provide more reward opportunity

6.3 Alignment Bonus Effect

For multiplicative variants, the bonus at $a > 0.7$ creates a “kink”:

- Below 0.7: gradient w.r.t. a is n (or 10η)
- Above 0.7: gradient w.r.t. a is $6n$ (or 60η)

This $6\times$ steeper gradient strongly incentivizes achieving $a > 0.7$, which correlates with clean lifts that don’t disturb the pile.

6.4 Failure Penalty Impact

- MR/MN use -3.0 : harsher penalty, may slow early exploration
- AR/AN use -0.5 : gentler penalty, allows more exploration but weaker negative signal

7 Recommendations

1. **For fixed pile sizes:** MR (original) converges well to ~ 100 reward
2. **For domain randomization:** MN recommended (strongest learning signal + pile-size agnostic)
3. **For interpretability:** AN provides cleanest semantics ($\eta + a$)
4. **For gentle exploration:** AR/AN with smaller failure penalty

8 Next Steps

Run standardized inference evaluation on all trained checkpoints measuring reward-agnostic metrics:

- Logs grasped per episode
- Grasp success rate (% of grasps with $n > 0$)
- Average alignment
- Pile clearing percentage

9 Appendix: Gym Task IDs

Without Domain Randomization

Isaac-Crane-Full-MR-v0	# Multiplicative Raw (original)
Isaac-Crane-Full-MN-v0	# Multiplicative Normalized
Isaac-Crane-Full-AR-v0	# Additive Raw
Isaac-Crane-Full-AN-v0	# Additive Normalized

With Domain Randomization

Isaac-Crane-Full-DR-MR-v0
Isaac-Crane-Full-DR-MN-v0
Isaac-Crane-Full-DR-AR-v0
Isaac-Crane-Full-DR-AN-v0